Daily Expense Tracker

1. Application Description:

The application is a console-based Daily Expense Tracker developed using Java.
It helps students manage their daily budget while ensuring they always keep a minimum required amount for transportation or essential needs.

The app tracks:

- Daily spending
- Remaining money
- Weekly spending
- Weekly savings
- Estimated monthly savings

2. Main Logic Explanation:

The core logic of the program is based on:

- **Weekly Loop**
  A `for` loop runs for 7 days to simulate a full week.
- **Daily Budget Setup**
  Each day the user enters:
  - Day name
  - Daily budget
  - Minimum amount to keep
- **Shopping Loop**
  A `while` loop allows the user to keep buying items during the day.
- **Purchase Condition (Main Logic)**

  if (currentMoney - price >= minimumKeep)

  If true:
  - The purchase is approved
  - The price is deducted
  - Daily spending increases

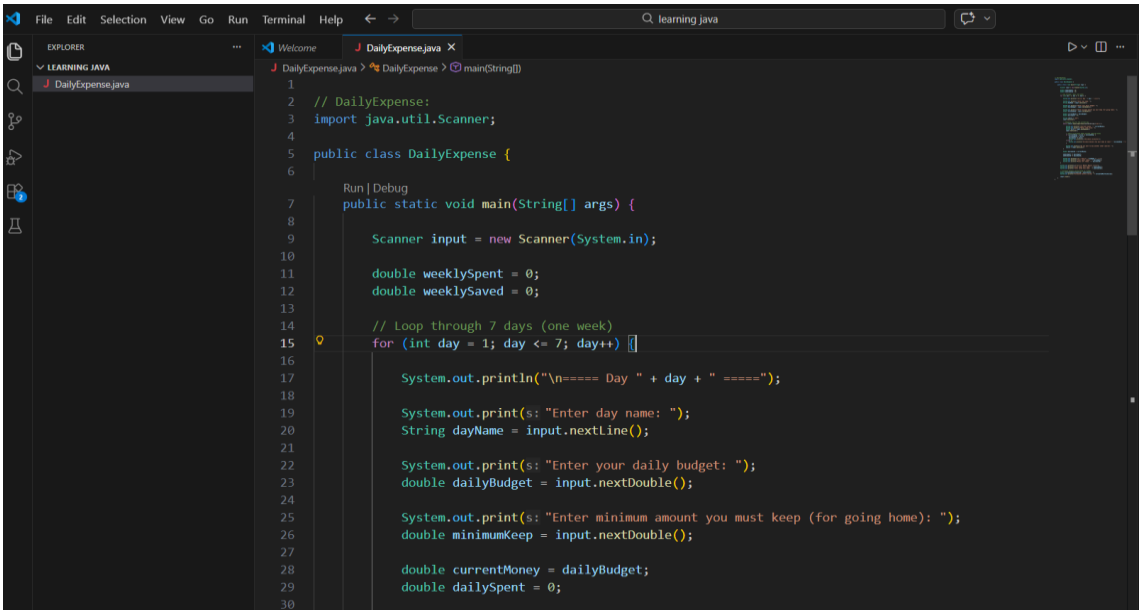
  If false:
  - The purchase is denied

- A warning message is shown

3. Main Function Code:

```
public static void main(String[] args) {
    Scanner input = new Scanner(System.in);

    double weeklySpent = 0;
    double weeklySaved = 0;

    for (int day = 1; day <= 7; day++) {

        System.out.println("\n===== Day " + day + " =====");

        System.out.print("Enter day name: ");
        String dayName = input.nextLine();

        System.out.print("Enter your daily budget: ");
        double dailyBudget = input.nextDouble();

        System.out.print("Enter minimum amount you must keep: ");
        double minimumKeep = input.nextDouble();

        double currentMoney = dailyBudget;
        double dailySpent = 0;

        String choice = "yes";
        input.nextLine();

        while (choice.equalsIgnoreCase("yes")) {

            System.out.println("\nCurrent money: " + currentMoney);
            System.out.print("Enter item price: ");
            double price = input.nextDouble();
            input.nextLine();

            if (currentMoney - price >= minimumKeep) {
                currentMoney -= price;
                dailySpent += price;
```

```
            System.out.println("Purchase successful");
        } else {
            System.out.println("Purchase denied");
        }

        System.out.print("Buy another item? (yes/no): ");
        choice = input.nextLine();
    }

    double dailySaved = currentMoney;

    weeklySpent += dailySpent;
    weeklySaved += dailySaved;

    System.out.println("Total spent today: " + dailySpent);
    System.out.println("Money left today: " + dailySaved);
  }

  System.out.println("Total spent this week: " + weeklySpent);
  System.out.println("Total saved this week: " + weeklySaved);

  double estimatedMonthlySaving = weeklySaved * 4;
  System.out.println("Estimated monthly saving: " + estimatedMonthlySaving);

  input.close();
}
```



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DailyExpense.java

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DailyExpense.java > DailyExpense > main(String[])

```java
5  public class DailyExpense {
7      public static void main(String[] args) {
31             String choice = "yes";
32             input.nextLine();
33
34             // Shopping loop for the current day
35             while (choice.equalsIgnoreCase(anotherString: "yes")) {
36
37                 System.out.println("\nCurrent money: " + currentMoney);
38                 System.out.print(s: "Enter item price: ");
39                 double price = input.nextDouble();
40                 input.nextLine();
41
42                 // Check if purchase keeps minimum required money
43                 if (currentMoney - price >= minimumKeep) {
44                     currentMoney -= price;
45                     dailySpent += price;
46                     System.out.println(x: "Purchase successful");
47                 } else {
48                     System.out.println("Purchase denied (You must keep at least " + minimumKeep + ")'
49                 }
50
51                 System.out.print(s: "Do you want to buy another item? (yes/no): ");
52                 choice = input.nextLine();
53             }
54
55             double dailySaved = currentMoney;
56
57             weeklySpent += dailySpent;
58             weeklySaved += dailySaved;
59
```

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```java
public class DailyExpense {
    public static void main(String[] args) {
            double dailySaved = currentMoney;

            weeklySpent += dailySpent;
            weeklySaved += dailySaved;

            System.out.println("\n--- End of " + dayName + " ---");
            System.out.println("Total spent today: " + dailySpent);
            System.out.println("Money left today: " + dailySaved);
        }

        System.out.println(x: "\n====== Weekly Report ======");
        System.out.println("Total spent this week: " + weeklySpent);
        System.out.println("Total saved this week: " + weeklySaved);

        // Estimate monthly saving based on 4 weeks
        double estimatedMonthlySaving = weeklySaved * 4;
        System.out.println("Estimated monthly saving: " + estimatedMonthlySaving);

        input.close();
    }
}
```